A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 26

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them **1 Built-in obsolescence:** products designed to have short lives / wear out / go out of fashion so customers upgrade **1** Disposable products (razors, lighters, pens) that must be replaced **1** Over-packaging to make products attractive **OR** increases wastes from manufacture and mineral extraction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1 Properties of the waste:** toxicity, flammability, degradability, radioactivity **1** Population density (collection costs) **1** Land availability for landfill **1** Availability of recycling technology / markets **1** Legislation / regulatory framework (landfill tax) **1** Public willingness to recycle / environmental awareness **1** Processing costs / household income **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** aerobic decomposition produces little methane – nutrients / organic matter returned to soil (circular economy) – reduces landfill volume **1 Disadvantage:** needs separation / collection of clean organic waste – odour / pests / pathogens if poorly managed – low-value product – not suitable for mixed waste **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
01.4	1 mark for the point below. **1** Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Effluent gases pass through a chamber containing electrically charged wires / plates **1** Smoke particles become (negatively) charged and are attracted to the oppositely charged plates where they collect / stick **1** Plates are shaken and the particles fall as fly ash which is collected (and may be used, eg in building materials) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **electrostatic precipitator** → charged plates that attract and collect particulates	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Effluent gases are forced to rotate / spin in a cylindrical chamber **1** The (denser) particles are thrown outwards to the chamber walls where they fall and collect – cleaned gases leave through a central pipe **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	1 mark for the point below. **1** The tar / volatile organic compounds that produce smoke on incomplete combustion are driven off, producing smokeless coal / coke **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Turbochargers deliver more air / oxygen to the combustion chamber **1** So combustion is more complete and fewer unburnt carbon particles / soot are produced (especially during acceleration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.5	**1 mark each** 1 mark for each of the 2 points below. **1** Smaller particles remain suspended in the atmosphere for longer so they are more likely to be inhaled **1** They penetrate deeper into the lungs / alveoli and are not cleared – may carry toxic / carcinogenic chemicals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - rivers are accessible over a large area with few natural contaminants	1

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1 Potable / public supply:** no pathogens, low toxins, good appearance / taste / smell (public health) **1 Industrial boiler / cooling water:** soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange – spray irrigation: low turbidity to avoid blocked pipes and low heavy metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) **1 Disadvantage:** high cost / energy for pumping – ecological impacts on the donor river – transfer of pollutants or invasive species – political conflict **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Low liposolubility and low persistence **Also accept** • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1 Bioaccumulation:** build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown **1 Biomagnification:** increase in concentration of the pollutant at successive trophic levels of a food chain **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **bioaccumulation** → a pollutant building up inside one organism over time • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** They are intended to be toxic (to the target species) **1** They are deliberately released / spread into the environment (rather than being an accidental or waste release) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 40-50 traps **1** Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Widely scattered results (large standard deviation) mean a single sample may not be representative **1** More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
05.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Same type and size of pitfall trap, set for the same length of time **1** Traps emptied at the same time of day / same weather conditions in both habitats **1** Same number of traps and same method of positioning (random or systematic) in each habitat **1** Other variables that could affect results (eg rainfall, temperature) monitored or controlled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **pitfall trap** → a sunken container that catches ground invertebrates • **random** → positions chosen using random numbers so the choice is not biased	3
05.4	**1 mark each** 1 mark for each of the 3 points below. **1** Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally **1** Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends **1** The interval between samples should match the rate at which the factor changes, found from a preliminary study **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals can only be exploited where they were deposited (no choice of location), unlike housing, roads or ports **1** Land take is larger than the mine void: buildings, access roads, overburden and spoil heaps, buffer zones – open-cast mines take far more land than deep mines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 5 points below. **1 Dust:** water sprays on roads / blasting areas make particles heavier and clump so they settle **1 Noise:** baffle mounds / embankments absorb and deflect – blasting at set, predictable daytime hours – quieter machinery **1 Turbid water:** sedimentation lagoons where water stands until solids settle before discharge **1 Leachate:** collect drainage water and pass through crushed limestone to neutralise acid and immobilise toxic metals **1 Amenity:** landscaping and tree planting to screen the mine – restoration plans agreed as a condition of planning permission **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Leachate** → polluted liquid draining out of waste	5
06.3	**1 mark each** 1 mark for each of the 3 points below. **1** Landscaping to reduce gradients, compaction, drainage to prevent waterlogging, adding soil / nutrients and planting vegetation whose roots bind the surface **1** At Aberfan a waterlogged coal-mine spoil heap became unstable and slid into the village **1** Killing 144 people, mainly children in the primary school **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogging** → soil pores filled with water so there is no air	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 139-142; AQA spec 3.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 6 points below. **1 Direct effect of abiotic change:** species near the edge of their range of tolerance (temperature, water) may die out locally, eg sundews if marshes dry, frogs if climate drier – beech (shallow roots) suffers in drought more than oak **1 Changes in timing:** earlier flowering / emergence may separate flowers from pollinators – plant toxins building up earlier may kill caterpillars – warmer winters disturb dormouse hibernation so they use fat reserves and starve **1 Biotic changes:** food supply, predators, disease vectors (mosquitoes / malaria spreading), pollinators and seed dispersers may change **1 Distribution changes:** species may colonise newly suitable areas (triton molluscs around the UK) but slow-growing species (trees) may disappear from old ranges faster than they colonise – suitable areas or biological corridors may not exist – human land use blocks movement **1 Population fragmentation:** rising seas or upward retreat to mountains isolate populations – divided gene pools increase inbreeding – small populations may not be viable – no repopulation from elsewhere **1** Some species benefit (bats: warmer winters, more insects) but wetter stormier weather reduces feeding time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **abiotic** → non-living / physical factors • **range of tolerance** → the range of conditions in which a species can survive • **Biotic** → living factors / effects of other organisms	6
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** All the species it depends on (food plants, pollinators, prey, mycorrhizae) will not move at the same speed **1** Less mobile species (plants, non-flying animals) cannot cross unsuitable areas without a biological corridor **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** The temperature difference between mid-latitude and polar air drives the jet streams – a smaller difference makes the jet stream slower and more meandering (Rossby waves) **1** Weather systems stall over an area, giving prolonged drought or rainfall / floods – unusually cold or warm air carried north or south **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 90-92; AQA spec 3.2.1.2.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) **1** It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
08.2	**Any 4 of 8** 1 mark each for any 4 of the 8 points below, in any order, maximum 4. **1** Physical state of the effluent (solution / solid) **1** Ocean currents / river flow **1** Effect of pH and oxygen on solubility **1** Bioaccumulation and food-chain concentration **1** Food sources and consumption by local people (eg shellfish, seaweed) **1** Half-lives of the isotopes **1** Meteorological conditions for any gaseous release **1** Nature of the environment - geology, vegetation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain	4
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring focuses on the sites most at risk of contamination **1** No need to sample sites that were never likely to be polluted – sampling checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2

Do not write outside the box

Question	Marking guidance	Marks
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** It assesses the risk to the people most exposed because of their lifestyle before any health effects occur **1** If their risk is acceptably low everyone else has a lower risk – if unacceptable, emissions are reduced to prevent future exposure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Critical Group** → the people most at risk / most heavily exposed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
09.1	**1 mark each** 1 mark for each of the 2 points below. **1 Fission:** splitting the nuclei of large atoms (U-235, Pu-239) when hit by neutrons **1 Fusion:** joining the nuclei of small atoms (deuterium and tritium - isotopes of hydrogen) to form helium, releasing a neutron and energy - the energy source of stars **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Deuterium (hydrogen-2) extracted from (sea)water **1** Tritium (hydrogen-3) produced by neutron bombardment of lithium (a lithium blanket around the reactor) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
09.3	**1 mark each** 1 mark for each of the 3 points below. **1** Very high temperature turns hydrogen into a plasma (electrons stripped away) and gives nuclei enough kinetic energy to overcome the repulsion between positive nuclei and collide **1** Vacuum so the plasma is not cooled by air **1** Magnetic field holds the plasma centrally so it does not touch the container walls and cool **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
09.4	**1 mark each** 1 mark for each of the 3 points below. **1** Fuel (deuterium / lithium) is abundant / effectively unlimited **1** Much less radioactive waste (no long-lived fission products) – no chain reaction / meltdown risk – no weapons material **1** Not yet achieved sustained net energy output: reactors (JET) use more energy than they release – ITER aims for 500 MW out from 50 MW in – laser fusion also experimental – commercial viability decades away **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 195-196; AQA spec 3.3.4.1.	
10.1	**1 mark each** 1 mark for each of the 3 points below. **1** Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) **1** They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone **1** The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
10.2	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Provides carbon, oxygen, hydrogen and nitrogen (as N_2, O_2, CO_2, H_2O) for biological molecules **1** Delays the escape of infrared energy, keeping the surface warm **1** Distributes heat and water vapour by winds / ocean currents **1** Provides atmospheric pressure so liquid water can exist **1** Deflects damaging radiation of the solar wind in the upper atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
10.3	1 mark for the point below. **1** Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **stratified** → sampling each area or group in proportion to its size	1
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Troposphere:** enhanced greenhouse effect - anthropogenic CO_2 (fossil fuels, deforestation), methane (rice, livestock, landfill), N_2O, CFCs, tropospheric ozone – more infrared absorbed and re-emitted – aerosols and smoke increase albedo / scatter light – land-use change alters surface albedo and evapotranspiration • **Stratosphere:** CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium – less UV absorbed in the stratosphere, more reaches the surface – stratospheric cooling • **Consequences:** changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions – increased UV damage to organisms • **Difficulties of prediction:** linked processes where one change alters others – feedback (positive and negative) and tipping points – time lags (ocean heat capacity, long residence times of CFCs and CO_2) – natural variability (solar output, volcanoes, Milankovitch cycles) – incomplete understanding of atmospheric processes – limited historical data so proxies needed – models with different assumptions • **Evaluation:** some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain – precautionary principle • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **enhanced greenhouse effect** → extra warming caused by added greenhouse gases • **anthropogenic** → human-caused / caused by human activity • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	
	Model answer - Level 3 (8 marks)	
	In the troposphere human activity enhances the greenhouse effect: fossil fuels and deforestation add CO_2, rice, livestock and landfill add methane, fertilisers add nitrous oxide, and CFCs and tropospheric ozone are new absorbers. More outgoing infrared is absorbed and re-emitted, warming the surface, while smoke and aerosols scatter sunlight and land-use change alters albedo and evapotranspiration. In the stratosphere, CFCs, halons and HCFCs release chlorine and bromine that catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium so that less UV is absorbed there and more reaches the surface, cooling the stratosphere as well.	
	The consequences run through every system the atmosphere's energy drives: climate, thermohaline circulation, the hydrological cycle, sea level and species distributions, plus UV damage to organisms. Predicting them is hard because the processes are linked - one change alters others - and because feedback can amplify (albedo loss, methane release) or damp (cloud cover, photosynthesis) the effect, sometimes past tipping points. Time lags from ocean heat capacity and the long residence times of CO_2 and CFCs, natural variability from solar output, volcanoes and orbital cycles, sparse historical data replaced by proxies, and models built on different assumptions all add uncertainty.	
	Some things are nevertheless well established: the physics of infrared absorption, the chemistry of ozone loss, and the success of the Montreal Protocol. What is uncertain is the scale and timing of regional impacts, which is exactly why the precautionary principle, not certainty, should guide action.	
	Weaker answer - Level 1 (3 marks)	
	Humans release greenhouse gases like carbon dioxide from burning fossil fuels into the troposphere, which traps more heat and causes global warming. CFCs go into the stratosphere and destroy the ozone layer, so more UV reaches the Earth. It is hard to predict what will happen because the atmosphere is complicated and scientists do not know everything about it. There could be feedback effects that make it worse.	
	What makes the difference	
	• Level 1 gives one cause per layer; Level 3 lists several gases and processes and explains the mechanism (absorption and re-emission; catalytic destruction shifting an equilibrium).	
	• 'Difficulties of prediction' is answered in Level 3 with specific reasons (linked processes, feedback, time lags, natural variability, proxy data, models); Level 1 says 'complicated'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 evaluates - separating what is certain from what is not - and links this to the precautionary principle.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	
11.1	**Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Historical data limitations – proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • **Satellite sensors:** visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring – passing through cloud • **Sensor carriers:** ROVs / drones, AUVs, balloons, aircraft, animals – data recorders for abiotic factors – GPS tracking of floats and sampling sites – Argo floats and ocean monitoring • Climate models and their assumptions – monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • **Evaluation:** improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why non-renewable:** formed too slowly – geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock – Lasky's principle – cut-off ore grade and its drivers • **Extraction technologies:** mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • **Reducing demand:** recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • **Environmental limits:** impacts of mining and their control – energy costs rising with lower grades – conflicts	25
	• **Judgement:** technology can extend supplies substantially but only within energy and environmental constraints – circular economy essential *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	

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